Cross-Domain Few-Shot Learning Based on Graph Convolution Contrast for Hyperspectral Image Classification

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***Abstract*— Training a deep-learning classifier notoriously requires hundreds of labeled samples at least. Many practical hyperspectral image (HSI) scenarios suffer from a substantial cost associated with obtaining a number of labeled samples. Few-shot learning (FSL), which can realize accurate classification with prior knowledge and limited supervisory experience, has demonstrated superior performance in the HSI classification. However, previous few-shot classification algorithms assume that the training and testing data are distributed in the same domains, which is a stringent assumption in realistic applications. To alleviate this limitation, we propose a cross-domain FSL based on graph convolution contrast (GCC-FSL). The proposed method leverages cross-domain learning to acquire transferable knowledge from the source domain for classifying samples in the target domain. Specifically, a positive and negative pairs module is designed for constructing positive and negative pairs by matching the class prototypes of the target domain with those of the source domain, which aligns the data distribution of the source and target domains. In addition, a graph convolution contrast (GCC) module is proposed for extracting global graph-structure information of HSI to improve the ability of feature expression and constructing a graph-contrast loss to solve a domain-shift problem. Finally, a multiscale feature extraction network is designed to expand convolutional receptive fields through feature reuse and increase information interaction for fine-grained feature extraction. The experimental results demonstrate the improved performance for the proposed FSL framework relative to both state-of-the-art convolutional neural network (CNN)-based methods as well as other few-shot techniques. The source code of this method can be found at https://github.com/JieW-ww/GCC-FSL.**

***Index Terms*— Contrastive learning, few-shot learning (FSL), graph convolution (GC), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL image (HSI) obtained by hyperspectral sensor is a 3-D data cube consisting of hundreds of spectral channels, for each pixel, each of which contains the values that correspond to the detailed spectrum of reflected light [1], [2]. HSI provides the potential for precise feature identification and has been widely applied in both military and civilian applications, such as agricultural detection, geological exploration, and security control (e.g., [3], [4], [5]).

In recent years, deep-learning-based methods have been widely used in HSI. Convolutional neural networks (CNNs) [6], as the mainstream algorithms of deep learning, possess remarkable capability in automatically learning nonlinear and hierarchical representations, thereby demonstrating excellent performance in HSI classification. Slavkovikj et al. [7] proposed a classification method based on 1D-CNN to extract deep spectral features in HSI. A 2D-CNN is used in [8] to acquire spatial features of HSI. Chen et al. [9] designed a framework based on 3D-CNN to extract the jointly spatial–spectral features for HSI classification. Zheng et al. [10] incorporated continuous 3D-CNN–2D-CNN and covariance pooling, proposing a mixed CNN (MCNN) to extract more discriminative spectral–spatial features and improve the performance of HSI classification. Li et al. [11] designed a neural network based on 3D-CNN that accurately classifies HSI by adding convolution layer in the spectral dimension. The classification accuracies of CNN-based models improve significantly as the network deepens, but excessively deep networks may suffer from gradient vanishing and exploding. To address this issue, residual structure [12] is incorporated into CNN-based models for HSI classification. Zhong et al. [13] proposed a spectral–spatial residual network (SSRN), which makes full use of shallow features and deep features by using residual structure to improve classification accuracy. Zhang et al. [14] proposed an efficient spectral partitioning residual network (SPRN), which utilizes the SP strategy to divide input patches along the spectral dimensions and constructs multiple parallel 2D-CNNs separately to extract the spectral features of HSI. The residual block in SPRN is improved by incorporating additional branches, which helps alleviate the problems of gradient vanishing and exploding. The preceding approaches for HSI classification perform well, but one drawback is that they require a lot of labeled data to train the network. Due to the time-consuming and expensive of data collection and labeling, the number of training data is limited in real HSI applications, which affects the classification

Manuscript received 14 November 2023; revised 16 December 2023; accepted 5 January 2024. Date of publication 10 January 2024; date of current version 23 January 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62201043 and in part by the China Postdoctoral Science Foundation under Grant 2021M700441. *(Corresponding author: Jinxin Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3352093

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results of traditional deep-learning-based methods. Thus, it is a challenge to achieve high-precision classification using a small number of labeled samples.

Inspired by the human visual recognition system, the researchers focused on the HSI classification methods based on few-shot learning (FSL) [15], [16], which has good recognition ability with limited samples. This model utilizes relational learning to calculate the similarity of spatial–spectral information from HSI and achieves accurate classification by expanding the variability of different classes. Liu et al. [17] proposed a deep FSL (DFSL) method, which utilizes a deep residual network to extract spatial–spectral features from the input dataset to generate a feature distribution space. Furthermore, the network is trained through episodes to learn the metric space, which ensures the proximity of samples from the same class and the distance of samples from different classes. Later, research points out that traditional FSL methods assume that the training and testing tasks are from the same domain. In real-word applications, the process of data collection from both training and testing tasks is inevitably influenced by various factors, such as sensor nonlinearity, seasonal, or weather effects [18], [19], [20]. These factors lead to differences in spectral information for the same target between training and testing tasks, which degrades the classification accuracy. To overcome the above limitation, deep cross-domain FSL (DCFSL) [21] was proposed for HSI classification, which uses meta-learning and domain adaptation to overcome domain shift while simultaneously enhancing knowledge transfer between source and target domains. Wang et al. [22] proposed a cross-domain FSL HSI classification method, optimizing feature extraction for discriminative and domain-invariant representations, and enhancing interclass decision boundaries through class-wise attention and asymmetric domain adversarial training. On a similar note, Xi et al. [23] introduced FSL with class-covariance metric (CMFSL). This method learns domain-invariant class representations by conducting interactive training between the base and novel classes and employing the Mahalanobis distance measurement instead of an extra classifier.

Although the above cross-domain learning methods mitigate the effect of domain shift to a certain extent, they only consider aligning the feature distribution between the source and target domains. However, this alignment may fail to effectively align some classes with significant differences between the source and target domains, thereby impacting the model's generalization ability. Contrastive learning [24], [25] usually focuses more on sample-level alignment, which can help retain more detailed information and reduce unnecessary feature distribution alignment, thus greatly alleviating the domain-shift problem. As an unsupervised learning, contrastive learning can learn semantic representation from unlabeled data alternative to supervised learning and has been gradually applied to the HSI classification. Zhang et al. [26] proposed a cross-domain few-shot contrastive learning for HSI classification, aiming to learn more discriminative representation for HSI that can conquer the obstacles from high interclass similarity and large intraclass variance. Hou et al. [27] proposed a novel HSI classification network based on contrast learning. First, they pretrain the network in the way of self-supervised learning and employ a large amount of unlabeled samples for contrastive learning. Then, in order to improve the classification performance, a small amount of labeled samples are used to fine-tune the extracted features. Liu et al. [28] proposed a refined prototypical contrastive learning network for FSL (RPCL-FSL), which incorporates supervised contrastive learning and FSL into an end-to-end network to perform HSI classification with small size of labeled samples. Although contrastive learning-based methods have been proved effective for HSI classification, even cross-domain HSI classification, there are still some problems: 1) achieving feature distribution alignment or class-level alignment between the source and target domains while overlooking the issue of negative transfer resulting from significant differences between the two domains and 2) utilizing conventional convolutional structures for extracting information from HSI while neglecting the inherent graph-structure information of HSI. In general, HSI can be characterized as irregular graph-structure data, where each pixel is considered as a vertex in the graph, and the relationships between the vertices are represented as edges in the graph [29], [30]. By converting the similarity and connectivity relationships among pixels into a graph structure, spatial and spectral features can be extracted for HSI classification task. For instance, dual graph cross-domain FSL [31] and graph information aggregation cross-domain FSL (Gia-CFSL) [32] have been proposed to demonstrate the effectiveness of graph-structure information. Therefore, it is necessary to combine contrast learning with graph-structure information for cross-domain FSL classification for HSI.

To address the limitations of the aforementioned methods, in this article, a cross-domain FSL based on graph convolution contrast (GCC-FSL) for HSI classification is proposed. For cross-domain FSL classification, how to match source-domain samples to the target-domain samples is the key to make full use of source-domain labeled samples. To address this, we introduce the concept of contrastive learning and design a module for constructing positive and negative pairs to determine the correspondence of sample pairs from source and target domains through distance measurement, thereby achieving alignment of feature distribution spaces at the sample level. Simultaneously, we employ a contrastive loss function to optimize model parameters, enhancing the similarity of positive pairs while reducing the similarity of negative pairs. For another issue, how to capture the geometric variations of different land cover classes is crucial for HSI feature expression. Especially for cross-domain classification, high-level graph features are conducive to expressing graph-structure information of source- and target-domain datasets. We devise a graph convolution contrast (GCC) strategy to measure graph-structure information and optimize parameters by employing a contrast learning loss function, aiming to elevate the similarity of positive pairs and diminish the similarity of negative pairs. This approach facilitates achieving sample-level alignment in the feature distribution space, thereby alleviating the issue of domain shift.

In summary, the contributions of this article are listed as follows.

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1) A GCC-FSL algorithm is proposed to address the cross-domain few-shot problem for HSI classification. The proposed method effectively converts the similarity and connectivity relationships of samples into a graph structure for extracting domain-invariant features.
2) In GCC-FSL, the positive and negative pairs' construction module is utilized to match the class prototypes between source and target domains, which aims to alleviate the domain shift between two domains.
3) To effectively extract graph-structure information from sample pairs, a GCC module is introduced, which is beneficial to feature expression and further addresses the domain-shift problem.
4) Experimental results reveal that GCC-FSL not only outperforms competing approaches but also provides highly stable performance even with a limited size of the training set.

The rest of this article is organized as follows. Section II provides a detailed description of the relevant theories on cross-domain FSL and graph convolution (GC). Section III describes the proposed GCC-FSL and its constituents. Section IV discusses the experimental results and analyzes the ablation study. Finally, we make concluding remarks in Section V.

II. Related Work

A. Cross-Domain Few-Shot Learning

Cross-domain FSL aims to accomplish the image classification task using a few labeled samples in the dataset (target domain) with a large number of labeled samples in other datasets (source domain), and meta-learning-based cross-domain FSL methods are widely used. Cross-domain FSL uses both source- and target-domain samples to train the model, utilizing transferable meta-knowledge from the source domain to facilitate accurate target-domain classification. In few-shot classification task, the source-domain dataset $\mathbf{D}_s$ contains $\mathbf{C}_s$ classes and the target-domain dataset $\mathbf{D}_t$ contains $\mathbf{C}_t$ classes. $\mathbf{C}_s$ and $\mathbf{C}_t$ represent the number of classes in the source and target domains, respectively. The number of classes in the source domain should be equal or more than the number of classes in the target domain. According to whether the data are labeled, the target-domain dataset can further split into two parts: few-shot dataset $\mathbf{D}_f$ with labeled data and test dataset $\mathbf{D}_q$ with unlabeled data, and $\mathbf{D}_t = \mathbf{D}_f \cup \mathbf{D}_q$.

During the training process of FSL, the sampled mini-batches called episodes are employed to conduct FSL tasks. Consider these episodes as FSL tasks and we sample them randomly from $\mathbf{D}_s$ or $\mathbf{D}_t$. FSL is accomplished by alternating training on source-domain tasks with target-domain tasks. Taking the source-domain training task as an example, first, $\mathbf{m}$ samples are randomly selected from $\mathbf{D}_s$ to form a support set $\mathbf{S}_s = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{C_s \times m}$, and then, $\mathbf{n}$ samples are randomly selected from $\mathbf{D}_s$ to form a query set $\mathbf{Q}_s = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{C_s \times n}$. In the training phase, the support set is used to train the model, while the query set is used to calculate the loss function to update the model parameters. The source-domain tasks are taken similar to the target-domain tasks. It is worth noting that the number of samples is few, and it is difficult to generate effective support and query sets. We add random Gaussian noise to expand the training sample size, inspired by [17] and [21].

B. Graph Convolution

A graph is a nonlinear data structure composed of multiple vertices connected by edges, each of which represents the similarity between two individual vertices. Therefore, it is commonly used for one-to-many correspondence in non-Euclidean spaces. GC is a common operator for processing unstructured data that operate directly on a graph to extract high-level features by aggregating the neighborhood information of vertices [33], [34]. In HSI processing, the relationships between spectral features can be represented using undirected graphs. Formally, $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ denotes an undirected graph, where $\mathcal{V}$ and $\mathcal{E}$ represent the set of vertices and the set of edges of the graph, respectively, and $\mathbf{A}$ represents the adjacency matrix. $\mathbf{A}_{i,j}$, which represents the weight value of the edge between graph vertex i and vertex j, can be defined in terms of the radial basis function (RBF)

$$\mathbf{A}_{i,j}^{\mathrm{RBF}} = \begin{cases} e^{-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2}, & \mathbf{x}_j \in \mathcal{N}_i^K \\ 0, & \text{else} \end{cases} \tag{1}$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ are connected in the graph, γ is a parameter controlling the spread of the RBF, and $\mathcal{N}_i^K$ is the set of K-nearest neighbors (K-NNs) to $\mathbf{x}_i \in \mathcal{V}$ (K is set to 1), that is, $\mathcal{N}_i^K$ is the set of $\mathbf{x}_j \in \mathcal{V}$ with the smallest spectral-domain Euclidean distance, where the spectral-domain Euclidean distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ is defined as $d(\mathbf{x}_i, \mathbf{x}_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2$; see [35] for more detail.

Thus, the adjacency matrix used to update the GC can be defined as

$$\mathbf{A} = \mathbf{D}^{-1/2}\left(\mathbf{A}_{i,j}^{\mathrm{RBF}} + \mathbf{I}\right)\mathbf{D}^{-1/2} \tag{2}$$

where $\mathbf{D} = \sum_j \mathbf{A}_{i,j}$ is the diagonalized degree matrix and $\mathbf{I}$ is a unit matrix with the same size as that of graph vertices.

Using (2), the propagation rule of GC is

$$f_{\mathcal{G}}(\mathbf{\Theta}_1, \mathbf{\Theta}_2) = \sigma(\mathbf{\Theta}_2 \xi(\sigma(\mathbf{\Theta}_1 \mathbf{A}))\mathbf{A}) \tag{3}$$

where $\xi(\cdot)$ represents the dropout layer and $\sigma(\cdot)$ represents the ReLU activation function. $\mathbf{\Theta}_1$ and $\mathbf{\Theta}_2$ represent the trainable model parameters in the GC.

III. Methodology

In this section, we describe the structure of GCC-FSL, an overview of which is shown in Fig. 1. The network is composed of feature extraction network, positive and negative pairs' construction module, and GCC module. For feature extraction, spatial–spectral features are extracted by a network based on 3D-CNN, the residual structure of which can expand convolutional receptive fields by feature reuse and increase information interaction for fine-grained feature extraction. The positive and negative pairs' construction module constructs the feature distribution space of positive and negative sample alignment for the source and target samples, improving the

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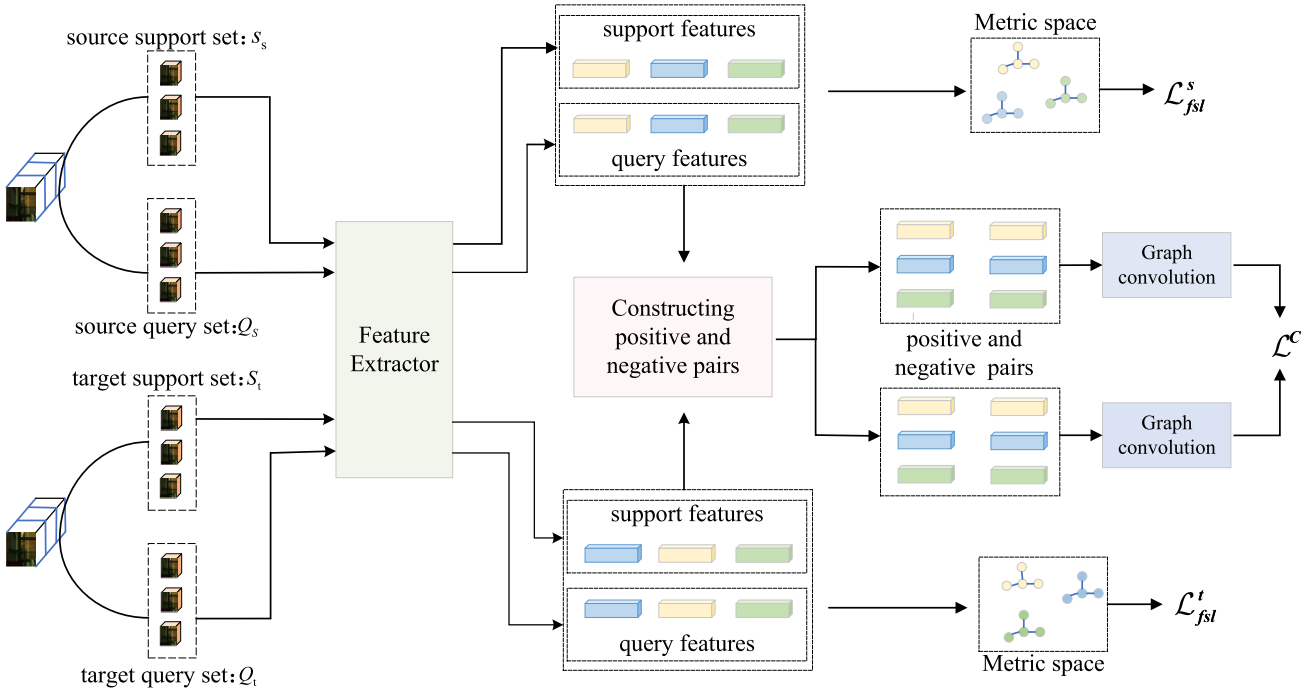


Fig. 1. Overall framework of GCC-FSL algorithm in the training phase.

ability of feature representation for few-shot cross-domain HSI classification. Moreover, we designed a GCC module to extract graph-structure information from positive and negative pairs and combined with contrast learning to reduce the distance between positive pairs and increase the distance between negative pairs, so as to alleviate the domain-shift problem. In the following, we describe each of the components in detail, starting with the overall framework in Section III-A. Also, a detailed description of feature extraction module and positive and negative pairs' construction module are presented in Sections III-B and III-C, respectively. In addition, we describe a GCC module in Section III-D.

A. GCC-FSL Framework

The GCC-FSL algorithm adheres to the widely adopted FSL framework in constructing support and query sets for model training. During the training phase, a certain amount of labeled samples from the source domain are utilized to assist in training the samples from the target domain. In the testing phase, the labeled samples from the target domain are employed to train the classifier.

In the training phase, FSL creates a support set $\mathbf{S}_s = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{C_s \times m}$, and a query set $\mathbf{Q}_s = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{C_s \times n}$ for the source dataset while creating a support set $\mathbf{S}_t = \{(\mathbf{x}_j, \mathbf{y}_j)\}_{j=1}^{C_t \times m}$ and a query set $\mathbf{Q}_t = \{(\mathbf{x}_j, \mathbf{y}_j)\}_{j=1}^{C_t \times n}$ for the target dataset. The original feature maps of the source and target data are transformed into a feature distribution space through the feature extraction network, in which the labels of the query samples are selected by measuring the distance between a sample from the query set and another sample from the support set. Specifically, the proposed method utilizes the concept of prototype network [36]. It employs the mean value of the support-set features in the feature distribution space as the class prototypes. The query-set features are then computed by measuring their distance to each class prototype. Subsequently, the classification probability of each class is obtained using Softmax. Finally, the labels of the query-set samples are determined as the classes with the highest classification probability. As a result, the classification loss function between the query set and the support set in the source domain is defined as

$$\mathcal{L}^s_{\text{fsl}} = -\log p_\phi(y = k|x) \tag{4}$$

where $\mathbf{p}_\phi$ is the integrated prediction of the classifier. The class distribution for a query sample x is

$$p_\phi(y = k|x) = \frac{\exp(-d(f_\phi(x), c_k))}{\sum_{k=1}^{C_t} \exp(-d(f_\phi(x), c_k))} \tag{5}$$

where c_k represents the prototype representation of the kth class, $f_\phi(x)$ represents the features of the query set, and $d(f_\phi(x), c_k)$ represents the Euclidean distance function.

Similarly, the classification loss of target-domain episode can be defined as

$$\mathcal{L}^t_{\text{fsl}} = -\log p_\phi(y = k|x). \tag{6}$$

During the FSL training iteration, both source- and target-domain samples are jointly utilized for classification loss to facilitate the deep-learning model's adaptation to data from both domains. The loss function of FSL is

$$\mathcal{L}_{\text{fsl}} = \mathcal{L}^t_{\text{fsl}} + \mathcal{L}^s_{\text{fsl}}. \tag{7}$$

In the training process, the normalized temperature-scaled cross-entropy loss (NT-Xent loss) function $\mathcal{L}^C$ [37] is added to measure the similarity of positive pairs between source domain and target domain by minimizing their distance while

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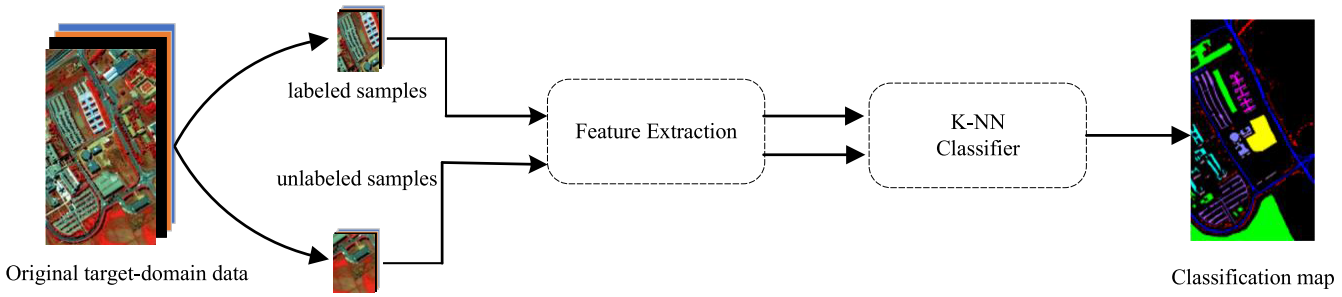


Fig. 2. Overall framework of GCC-FSL algorithm in the testing phase.

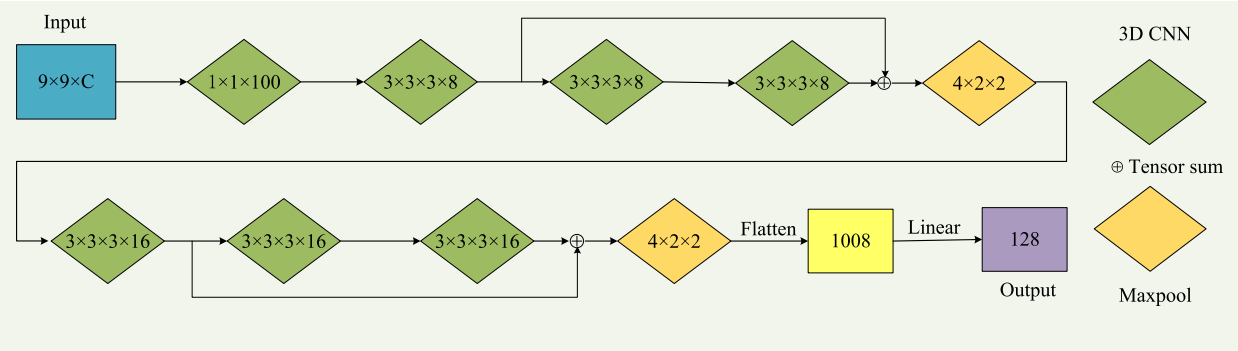


Fig. 3. Feature extraction network.

decreasing the similarity of negative pairs between source domain and target domain by maximizing their distance.

Therefore, in the training phase, the loss function of GCC-FSL can be described as

$$\mathcal{L} = \mathcal{L}_{\text{fsl}} + \lambda \mathcal{L}^{C} \tag{8}$$

where λ is the regularization parameter.

In the testing phase, embedding features are extracted from target-domain data and used to train a K-NN classifier (K is set to 1). Later, the trained classifier is applied to classify the unlabeled samples in the target domain and finally obtain the classification map of the target-domain data. The testing process of the GCC-FSL network is shown in Fig. 2.

TABLE I
NETWORK LAYER SIZES FOR THE FEATURE EXTRACTION NETWORK

Layer Name	Input Size	Filter Size	Output Size
Conv1	$9 \times 9 \times C$	$1 \times 1 \times 100$	$9 \times 9 \times 100$
Conv2	$9 \times 9 \times 100 \times 1$	$3 \times 3 \times 3 \times 8$	$9 \times 9 \times 100 \times 8$
Conv3	$9 \times 9 \times 9 \times 8$	$3 \times 3 \times 3 \times 8$	$9 \times 9 \times 100 \times 8$
Conv4	$9 \times 9 \times 9 \times 8$	$3 \times 3 \times 3 \times 8$	$9 \times 9 \times 100 \times 8$
Conv2 + Conv4			
Maxpool	$9 \times 9 \times 100 \times 8$	$4 \times 2 \times 2$	$5 \times 5 \times 28 \times 8$
Conv5	$5 \times 5 \times 28 \times 8$	$3 \times 3 \times 3 \times 16$	$5 \times 5 \times 28 \times 16$
Conv6	$5 \times 5 \times 28 \times 16$	$3 \times 3 \times 3 \times 16$	$5 \times 5 \times 28 \times 16$
Conv7	$5 \times 5 \times 28 \times 16$	$3 \times 3 \times 3 \times 16$	$5 \times 5 \times 28 \times 16$
Conv2 + Conv4			
Maxpool	$5 \times 5 \times 25 \times 16$	$4 \times 2 \times 2$	$3 \times 3 \times 7 \times 16$
Flatten	$3 \times 3 \times 7 \times 16$		1008
Linear	1008		128

B. Feature Extraction Based on 3D-CNN

In this article, feature extraction is performed using a 3D-CNN, which is able to simultaneously perform convolutional operations in both the spectral and spatial dimensions for obtaining discriminative features of HSI. For cross-domain FSL task, the feature extraction network is oriented to both source-domain data and target-domain data. Since the target-domain data and the source-domain data are often acquired by different types of sensors, their spectral dimensions are inconsistent. Therefore, it is necessary to establish a mapping layer to map them into the same dimension prior to cross-domain feature extraction. Specifically, a 1×1 convolution is used to reduce the dimensionality of the source- and target-domain data and this convolution can merge the information between the channels, and then, the joint spatial–spectral information is extracted using a 3-D residual network structure, which reduces the problem of information loss in the convolution layer by superimposing the input information onto the processed information in the convolution layer through bypass branches. The pooling layer is then used to implement downsampling to speed up the computation and prevent overfitting. The structure of the feature extraction network is shown in Fig. 3. Table I provides a detailed summary of the proposed feature extraction network in terms of layer type as well as input and output feature-map sizes. In order to reduce computational complexity and the number of parameters, the output size of the final linear layer is set to 128, and the classification task is ultimately accomplished by feeding this output into a Softmax classifier.

C. Positive and Negative Pairs' Construction

In general, the similarity of adjacent class prototypes is higher, while the samples in these class prototypes are expected to possess high similarity. The target-domain samples

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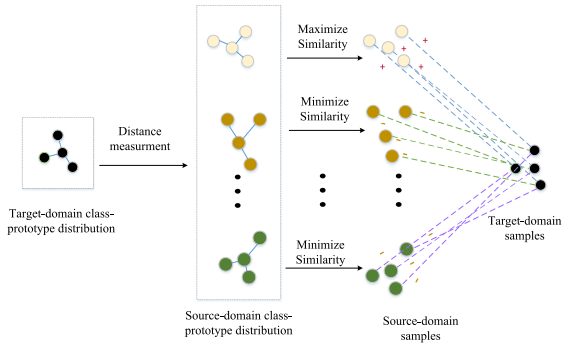


Fig. 4. Framework of positive and negative pairs' construction.

match the source-domain samples with higher similarity as positive pairs and the rest of the samples as negative pairs.

In order to establish positive and negative pairs between the source domain and the target domain, we employ class prototypes to represent classes in both domains. Subsequently, for each class prototype in the target domain, we systematically compare it with class prototypes in the source domain and select the source-domain class prototype that exhibits the closest distance (highest similarity) to pair them as positive pairs. Conversely, if such similarity criteria are not met, they are paired as negative pairs. A schematic illustrating the positive and negative pairs' construction module is shown in Fig. 4.

Specifically, the proposed method employs the randomly selected source- and target-domain samples and extracts features through a feature extractor f_θ and then measures the similarity of samples in each class by calculating the distance between the class prototype of the source domain (the mean value of support-set samples for each class in source domain) and the class prototype of the target domain (the mean value of support-set samples for each class in target domain). For example, if there are $\mathbf{C}_s$ and $\mathbf{C}_t$ classes in the source and target domains, respectively, then m samples are randomly drawn from each class of the source and target domains to form the support set S_p^s (p denotes the pth class of the source domain) for each class of the source domain and the support set S_q^t (q denotes the qth class of the target domain) for each class of the target domain. Using the mean value of samples as class prototypes $\mathbf{c}_p^s$ (source-domain class prototype) and $\mathbf{c}_q^t$ (target-domain class prototype), the similarity between $\mathbf{c}_p^s$ and $\mathbf{c}_q^t$ is calculated by the Euclidean distance.

The class prototype of class p in the source domain can be calculated by

$$c_p^s = \frac{1}{m} \sum_{x_i \in S_p^s} f_\theta(x_i), \quad p = (1, 2, \ldots, \mathbf{C_s}). \tag{9}$$

The class prototype of class q in the target domain can be calculated by

$$c_q^t = \frac{1}{m} \sum_{x_j \in S_q^t} f_\theta(x_j), \quad q = (1, 2, \ldots, \mathbf{C_t}). \tag{10}$$

The distance between source- and target-domain class prototypes can be calculated by

$$d\left(c_p^s, c_q^t\right) = \left\| c_p^s - c_q^t \right\|^2, \quad \begin{cases} p = (1, 2, \ldots, \mathbf{C_s}) \\ q = (1, 2, \ldots, \mathbf{C_t}). \end{cases} \tag{11}$$

By the calculation of (11), the classes belonging to the two class prototypes with the closest distance (the highest similarity) are defined as positive sample pairs, and the rest are defined as negative pairs.

D. Graph Convolution Contrast

To enhance the similarity of positive pairs and reduce the similarity of negative pairs, a GCC module is proposed. This module utilizes GC to extract high-level graph features from samples in both the source- and target-domain datasets, making full use of their graph-structure information. Also, contrast learning is utilized to improve the similarity of positive pairs and reduce the similarity of negative pairs in the feature distribution space. The architecture of the proposed GCC module is shown in Fig. 5.

The module is designed to find the samples with high similarity between target domain and source domain through introducing graph information into contrast learning, realizing few-shot HSI classification. After utilizing the positive and negative pairs' construction module to align source- and target-domain data, GC is employed to process these samples in the feature distribution space, fully exploiting the graph-structure information of HSI. Specifically, adjacent matrices are created for the source-domain samples and target-domain samples, and the matrices are used to obtain graph-structure features.

The feature $\hat{\mathbf{x}}_i$ of each sample in the source domain is obtained by GC. The sample $\mathbf{x}_j$ in the target domain is processed in the same way as the source-domain samples, and the feature $\hat{\mathbf{x}}_j$ of each sample in the target domain is extracted.

Given a graph G with N nodes, where the node set is defined as $V = 1, 2, \ldots, N$, for node i, there is a feature vector $\hat{\mathbf{x}}_i$ that represents the features of that node. X is the feature matrix, $X \in \mathbb{R}^{N \times F}$, where F is the dimension of the feature. A is the adjacency matrix, $A \in \mathbb{R}^{N \times N}$, which represents the connectivity relationships and weights between nodes. $H^{(0)}$ is the initial identity matrix. W is the weight matrix, which is used to learn the transformation of the node features.

The feature $\hat{\mathbf{x}}_i$ of each sample in the source domain can be calculated by

$$\hat{\mathbf{x}}_i = \sigma\left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(0)} W\right) \tag{12}$$

where $\tilde{A} = A + I$, I is the unit matrix, $\tilde{D}$ is the degree matrix, and σ is the activation function.

In order to enhance the similarity between positive pairs in both source and target domains while reducing the similarity between negative pairs, our proposed module employs the NT-Xent loss commonly used in contrastive learning to process the similarity relationship between the positive and negative pairs. The NT-Xent loss function is defined as

$$\mathcal{L}^C = -\log \frac{\exp\left(\text{sim}\left(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_j\right)/\tau\right)}{\sum_{m=1}^{2\,\text{N}} \exp\left(\text{sim}\left(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_m\right)/\tau\right)} (m \neq i) \tag{13}$$

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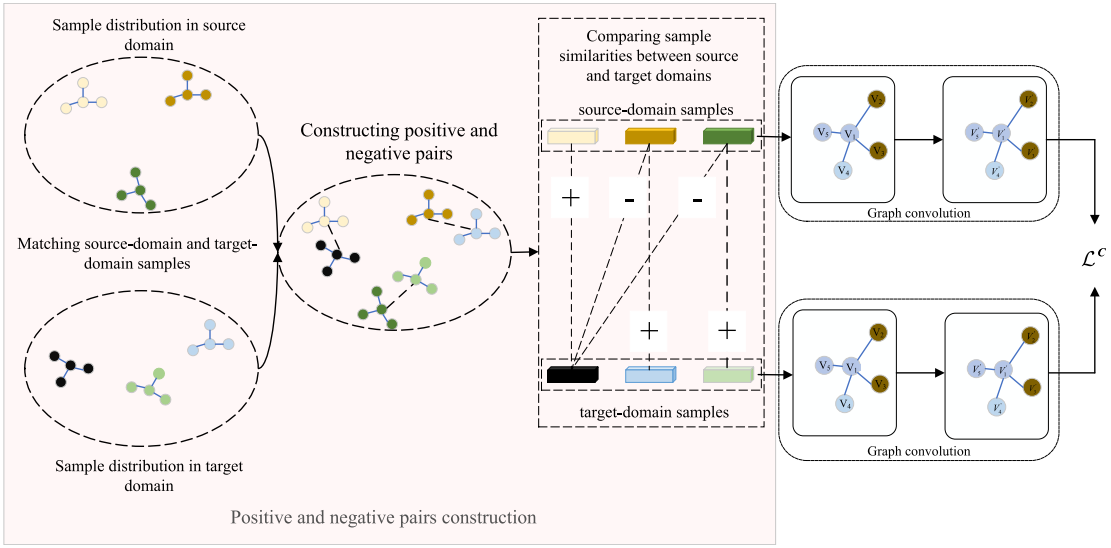


Fig. 5. Architecture of the proposed GCC.

TABLE II
CLASS NAME AND SAMPLE SIZE FOR THE CHIKUSEI DATASET

Class Number	Name	Number
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592

where the computation of target-domain feature $\hat{\mathbf{x}}_j$ is the same as $\hat{\mathbf{x}}_i$, N represents the number of positive and negative pairs in the feature distribution space, sim represents the cosine similarity, and τ represents the penalty factor for the negative pairs. In sum, the numerator of (13) represents the similarity of positive pairs and the denominator of it represents the similarity of negative pairs.

With the GCC module, the graph-structure features are extracted to form a feature distribution space. With the help of contrastive learning, the similar sample pairs in the feature distribution space are closer and the distances of dissimilar sample pairs are further away from each other. Thus, the domain-shift problem will be mitigated due to the enhancement of feature representation between the source and target domains.

IV. DATASETS AND EXPERIMENTAL SETUP

A. Experimental Datasets

To verify the effectiveness of the proposed GCC-FSL, in our experiments, the Chikusei dataset is used as the source-domain dataset, while the University of Pavia, Houston, and Hangzhou datasets are employed as the target-domain datasets.

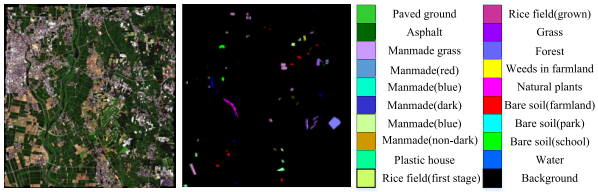


Fig. 6. (a) False-color image and (b) ground-truth map for the Chikusei dataset.

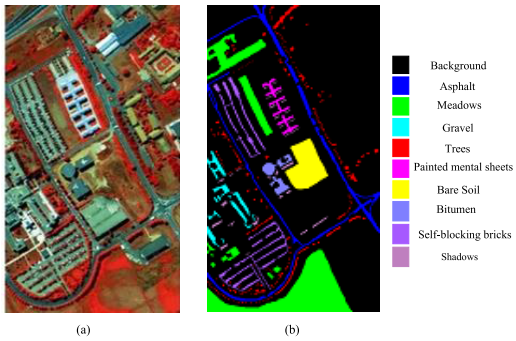


Fig. 7. (a) False-color image and (b) ground-truth map for the University of Pavia dataset.

1) Chikusei Dataset: It was taken by the Headwall Hyperspec-VNIR-C sensor in 2014 at Chikusei, Tsukushi, Japan. The dataset contains 128 wavelengths in the band range of 340–1018 nm, with a spatial size of 2517 × 2335 and a spatial resolution of 2.5 m. The dataset contains a total of 19 different classes, covering both urban and rural areas. The details of the Chikusei dataset are shown in Table II, and the false-color image and ground-truth map are shown in Fig. 6.

2) University of Pavia Dataset: The University of Pavia dataset was acquired by the Reflection Optical System Imaging Spectrometer (ROSIS) sensor in the city of Pavia, Italy, in 2003. The University of Pavia dataset contains nine classes. The dataset contains 103 usable spectral bands after removing 12 noise bands; the spatial size of the dataset is 610 × 340, while the spatial resolution is 1.3 m. Fig. 7 shows the

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TABLE III
CLASS NAME AND SAMPLE SIZE FOR THE UNIVERSITY OF PAVIA DATASET

Class Number	Name	Number
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal	1345
6	Bare soil	5029
7	Bitumen	1330
8	Self-blocking	3682
9	Shadows	947
	Total	42776

TABLE IV
CLASS NAME AND SAMPLE SIZE FOR THE HOUSTON DATASET

Class Number	Name	Number
1	Healthy grass	1251
2	Stressed grass	1254
3	Synthetic grass	697
4	Healthy grass	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking lot1	1233
13	Parking lot2	469
14	Tennis court	428
15	Running track	660
	Total	152029

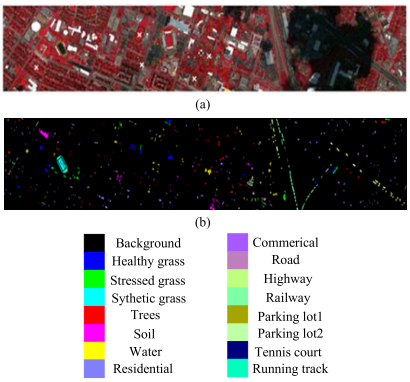


Fig. 8. (a) False-color image and (b) ground-truth map for the Houston dataset.

false-color image and ground-truth for the University of Pavia. The specific details of this dataset are shown in Table III.

3) Houston Dataset: The Houston dataset was collected by the Compact Airborne Spectrographic Imager (CASI) over the campus of the University of Houston and the neighboring urban area in June 2012. The dataset has a high spatial resolution of 349 × 1905 pixels. There are 15 classes, and the dataset is composed of 144 spectral bands covering the wavelength range from 380 to 1050 nm. The specific details of the Houston dataset are shown in Table IV, and a false-color image and ground-truth map are shown in Fig. 8.

4) Hangzhou Dataset: The NASA EO-1 Hyperion sensor obtained the Hangzhou dataset. There are 198 spectral bands after uncalibrated and noisy bands are removed. The spatial size of the Hangzhou dataset is 590 × 230 with 30-m spatial resolution. Three land cover classes listed in Table V

TABLE V
CLASS NAME AND SAMPLE SIZE FOR THE HANGZHOU DATASET

Class Number	Name	Number
1	Water	18043
2	Land/Building	77450
3	Plant	40207
	Total	135700

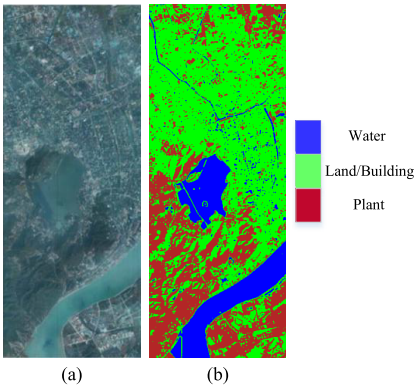


Fig. 9. (a) False-color image and (b) ground-truth map for Hangzhou dataset.

were selected for cross-domain classification in this work. Its false-color image and ground-truth map are shown in Fig. 9.

B. Hyperparametric Tuning

Since the selection of hyperparameters will affect the performance of the proposed GCC-FSL, we conducted experiments for hyperparameter tuning. The regularization parameters and the penalty factor for the negative pairs, i.e., λ and ξ, regarded as adjustable hyperparameters are selected from [0.001, 0.01, 0.1, 1] and [0.2, 0.4, 0.6], respectively. In the experimental data from different combinations of λ and ξ, the proposed GCC-FSL demonstrates the trend of classification accuracy (represented by OA%) on three datasets, as shown in Fig. 10. For the University of Pavia, Houston, and Hangzhou datasets, the optimal parameters are $\{\lambda = 0.01$ and $\xi = 0.2\}$, $\{\lambda = 0.1$ and $\xi = 0.4\}$, and $\{\lambda = 0.001$ and $\xi = 0.6\}$, respectively. The experimental results of the parameter tuning are shown in Fig. 10.

C. K-Nearest Neighbor Classifier Parameter Analysis

In the testing phase, we employed the K-NNs classifier for a classification task. In the K-NN classifier, the parameter K represents the number of nearest neighboring samples considered for classification. Therefore, we conducted experiments to select the parameter. As can be seen from Table VI, K is set to 1.

D. Parameter Settings

In this article, a $9 \times 9 \times N$ block is used to partition the original data as the input of the feature extraction network and a 1×1 convolution is used as the mapping layer to match the sizes of source and target data. All convolutional kernels are initialized using Xavier normalization [38]. The network uses the Adam optimizer [39] with cross-entropy loss, and the learning rate is 0.01 over 10 000 epochs. The number

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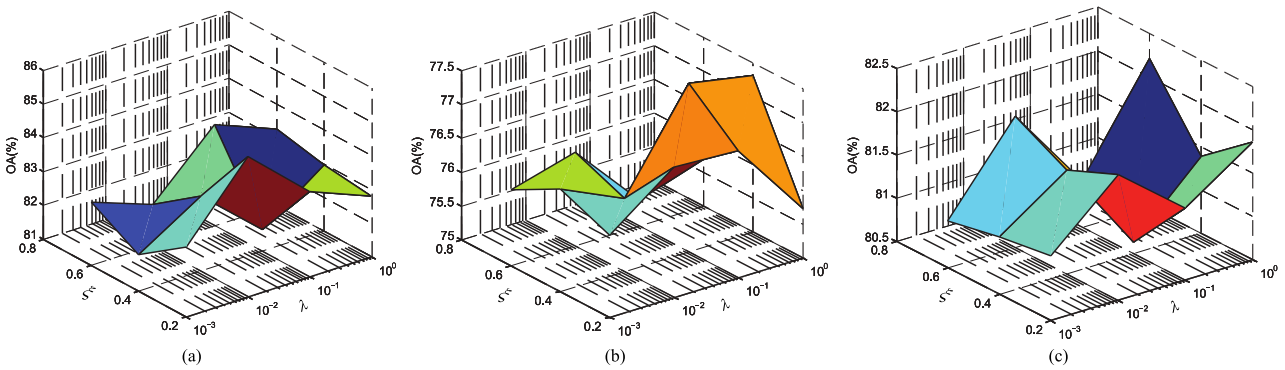


Fig. 10. Parameter tuning for the proposed GCC-FSL. (a) University of Pavia. (b) Houston. (c) Hangzhou.

TABLE VI
K-NN PARAMETER ANALYSIS

K		University of Pavia	Houston	Hangzhou
1	OA	84.85±1.63	77.50±0.99	82.21±0.71
	AA	84.02±1.79	79.83±1.04	82.90±1.78
	κ	80.12±2.13	75.71±1.06	69.19±1.53
2	OA	83.51±1.36	76.15±1.05	78.71±4.00
	AA	82.00±2.05	78.56±1.05	79.49±3.38
	κ	78.26±1.76	74.24±1.12	63.29±6.32
3	OA	84.13±3.34	76.29±0.86	79.04±4.12
	AA	84.06±2.33	78.78±0.75	80.21±2.95
	κ	79.30±4.22	74.39±0.91	64.05±6.40

of training iterations is set to 10 000. In the FSL training phase, the number of samples in the support set both in source and target domains is set to 1. In theory, the classification performance on target-domain data is expected to improve as the number of samples in the support set increases. The number of samples in the query set is set to 19.

V. RESULTS AND ANALYSIS

A. Comparison to State-of-the-Art Methods

To demonstrate the effectiveness of the proposed GCC-FSL algorithm, several methods are reported in this section, including traditional SVM [40], four state-of-the-art CNN-based methods (3D-CNN [11], SSRN [13], MCNN [10], and SPRN [14]), two FSL methods (DFSL + NN [17] and DFSL + SVM [17]), and three cross-domain FSL methods (DCFSL [21], CMFSL [23], and Gia-CFSL [32]). The 3D-CNN employs a single-branch cascade of conventional 3-D convolutions for feature extraction, followed by a fully connected layer for classification. SSRN is likewise a single-branch cascade of conventional 3-D convolutions but with added skip connections to form a residual network to enhance the expressiveness of the model. MCNN jointly uses 3-D and 2-D convolutions to extract spatial–spectral features and obtain second-order information from the feature map using covariance pooling techniques. SPRN deploys parallel 2-D residual CNNs on multiple partitions of spectral bands followed by feature fusion via pointwise convolution. We also compare the proposed method with two FSL methods (DFSL + NN and DFSL + SVM), which are successful cases of applying the FSL technique to the HSI classification. These methods utilize labeled samples from the source domain to acquire transferable knowledge that aids in classifying unlabeled samples from the target domain. Finally, we study three competing cross-domain FSL methods, including DCFSL, CMFSL, and Gia-CFSL. DCFSL utilizes two mapping layers and a spectral–spatial feature extraction network for FSL. In this network, a conditional adversarial domain adaptation strategy is adopted to alleviate the domain-shift problem. CMFSL proposed a class-covariance metric by using the Mahalanobis distance with task-adapted class-covariance estimation to improve the decision boundary of the nonlinear classifier. This improvement is achieved by considering the distribution of each class in the potential space, thereby enhancing few-shot classification accuracy. In addition, CMFSL incorporated a refinement module with spectral prior, effectively mitigating the impact of domain shift. Gia-CFSL proposed a graph information aggregation cross-domain FSL framework for HSI classification. This method employs a strategy of graph neural network information propagation and graph alignment. It considers the nonlocal spatial information with strong correspondence in both the feature and distribution levels to perform cross-domain graph alignment, thereby effectively mitigating domain shift.

The proposed method and all comparative experimental methods randomly employ five labeled samples for training, while the remaining samples are used for testing. The cross-domain methods, such as DCFSL [21], CMFSL [23], Gia-CFSL [32], and the proposed GCC-FSL, employ the Chikusei dataset as the source-domain data. Since the spectral dimensions of the source-domain data and the target-domain data were different, we applied 1×1 convolutions to perform dimensionality reduction on the source-domain data and the target-domain data for facilitating subsequent tasks.

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TABLE VII

CLASSIFICATION PERFORMANCE (IN PERCENT ±STD.DEV.) FOR THE UNIVERSITY OF PAVIA DATASET

	SVM [40]	3D-CNN [11]	SSRN [13]	MCNN [10]	SPRN [14]	DFSL+NN [17]	DFSL+SVM [17]	DCFSL [21]	CMFSL [23]	Gia-CFSL [32]	GCC-FSL
Asphalt	53.32	70.14	61.85	58.23	72.63	73.41	90.13	80.37	77.31	79.25	82.11
Meadows	78.14	62.43	69.82	80.31	66.85	82.85	83.02	85.49	85.54	86.62	90.43
Gravel	44.34	50.78	62.50	74.62	62.88	66.27	67.34	63.74	60.15	60.17	66.71
Trees	90.08	84.29	81.72	83.23	95.92	96.19	95.86	94.27	96.29	92.69	92.74
Painted metal	91.81	85.60	98.00	99.96	99.87	99.03	99.38	99.51	99.68	98.54	98.86
Bare soil	10.65	49.37	73.05	74.82	87.97	75.28	66.08	76.30	77.22	70.92	75.05
Bitumen	95.64	70.13	98.02	88.14	98.04	80.91	77.28	79.42	76.74	82.73	80.77
Self-blocking	52.03	74.66	77.88	55.63	74.66	65.52	61.09	59.94	74.91	78.75	71.61
Shadows	99.92	94.56	96.68	95.07	99.81	99.93	99.50	98.67	99.92	95.14	97.86
OA	64.74±1.79	65.81±1.17	74.51±1.49	75.23±1.25	75.52±0.74	79.97±1.67	81.09±1.68	81.52±3.50	82.39±1.00	82.53±3.02	**84.85±1.63**
AA	68.37±2.08	71.33±2.40	79.95±1.81	78.89±2.74	79.29±2.49	82.15±1.71	82.19±2.48	81.97±1.76	83.09±1.46	82.76±1.96	**84.02±1.79**
κ	54.28±2.12	57.27±1.71	67.95±1.53	68.25±1.62	69.64±1.01	74.13±2.04	74.02±2.89	76.10±4.06	77.21±1.76	77.23±3.63	**80.12±2.13**

TABLE VIII

CLASSIFICATION PERFORMANCE (IN PERCENT ±STD.DEV.) FOR THE HOUSTON DATASET

	SVM [40]	3D-CNN [11]	SSRN [13]	MCNN [10]	SPRN [14]	DFSL+NN [17]	DFSL+SVM [17]	DCFSL [21]	CMFSL [23]	Gia-CFSL [32]	GCC-FSL
Healthy grass	83.19	84.85	81.33	82.44	85.78	87.09	88.57	90.24	84.73	88.70	89.63
Stressed grass	55.96	71.81	84.28	73.17	85.70	79.29	77.79	80.21	94.10	81.43	80.10
Synthetic grass	98.20	93.88	99.88	93.81	99.91	96.29	95.92	98.27	96.10	96.21	96.34
Healthy grass	81.59	89.79	81.78	81.38	92.93	92.47	90.20	91.75	93.02	89.28	92.32
Soil	78.30	90.20	96.60	95.54	98.53	96.67	93.52	96.67	97.82	94.19	96.82
Water	84.65	70.68	87.56	92.14	86.75	78.56	72.06	83.75	82.66	76.44	77.71
Residential	29.15	53.03	75.93	46.25	77.88	71.67	64.51	68.63	52.08	61.25	69.42
Commercial	25.98	35.60	28.41	53.00	30.51	46.91	38.47	36.06	42.72	44.27	41.00
Road	77.96	45.42	66.58	46.65	64.49	53.18	52.86	64.54	64.00	63.62	65.50
Highway	3.51	49.08	47.81	68.07	43.49	58.34	65.78	63.22	58.37	64.93	72.33
Railway	58.18	62.85	65.07	68.55	62.46	57.79	62.59	56.65	67.87	62.75	64.82
Parking lot1	17.14	46.26	42.30	62.74	37.52	67.06	76.53	76.56	55.48	72.92	73.40
Parking lot2	9.48	40.24	74.57	92.91	93.06	87.91	89.66	90.99	95.10	89.76	88.18
Tennis court	93.51	87.49	99.81	100.00	100.00	93.66	91.63	87.14	98.70	92.98	95.31
Running Track	93.89	92.30	99.60	99.67	100.00	96.79	97.16	95.82	98.32	92.96	94.55
OA	55.84±3.08	65.76±2.35	71.63±1.87	72.56±0.92	73.35±0.97	74.69±1.13	74.55±1.10	75.92±2.89	75.16±1.43	75.56±1.20	**77.50±0.99**
AA	59.38±2.85	67.55±2.44	75.43±1.81	77.09±0.64	77.27±0.70	77.58±0.96	77.15±1.27	78.70±2.28	78.74±1.21	78.11±0.86	**79.83±1.04**
κ	52.54±3.27	63.01±2.53	69.40±2.01	70.41±0.98	70.90±1.31	72.67±1.22	72.51±1.20	73.98±2.59	73.21±1.52	73.60±1.28	**75.71±1.06**

TABLE IX

CLASSIFICATION PERFORMANCE (IN PERCENT ±STD.DEV.) FOR THE HANGZHOU DATASET

	SVM [40]	3D-CNN [11]	SSRN [13]	MCNN [10]	SPRN [14]	DFSL+NN [17]	DFSL+SVM [17]	DCFSL [21]	CMFSL [23]	Gia-CFSL [32]	GCC-FSL
Water	80.98	80.24	94.66	84.26	88.65	87.26	83.44	86.07	85.92	85.06	84.32
Land/Building	54.59	61.06	72.68	65.34	72.24	78.85	80.51	82.47	76.51	80.49	81.21
Plant	62.39	65.66	44.55	73.49	61.94	70.21	60.63	71.07	74.31	76.63	83.17
OA	60.41±1.72	65.24±1.11	67.27±1.77	70.27±0.95	71.37±1.54	77.41±1.66	75.01±2.03	79.57±3.80	77.84±2.13	79.95±3.63	**82.21±0.71**
AA	65.98±7.76	69.65±1.81	70.63±1.01	74.36±1.24	74.27±2.81	78.77±1.46	74.86±2.83	79.87±2.45	79.64±0.89	80.73±2.34	**82.90±1.78**
κ	36.95±4.74	44.11±1.43	43.39±2.54	51.27±1.02	50.75±4.05	60.67±2.69	55.45±3.79	64.11±5.92	62.09±2.46	65.29±3.44	**69.19±1.53**

TABLE X

ABLATION EXPERIMENT

		University of Pavia	Houston	Hangzhou
Network A	OA	83.02±1.88	75.39±2.38	78.03±2.72
	AA	83.13±1.94	79.05±1.98	80.08±3.46
	κ	77.78±2.35	73.44±2.52	62.75±2.19
	Parameters	0.185192M	0.185192M	0.185192M
	FLOPs	4.2739648G	4.2739648G	4.2739648G
Network B	OA	84.85±1.63	77.50±0.99	82.21±0.71
	AA	84.02±1.79	79.83±1.04	82.90±1.78
	κ	80.12±2.13	75.71±1.06	69.19±1.53
	Parameters	0.185192M	0.185192M	0.185192M
	FLOPs	4.2739648G	4.2739648G	4.2739648G

1) Quantitative Analysis: To gauge performance, we calculate overall accuracy (OA), average accuracy (AA), and Kappa (κ) coefficient. OA is the ratio of the number of correct prediction made by the network to the overall number of pixels in all test sets, while AA is the average value of the ratio of the correct prediction to the total size of each class. The calculation of the κ coefficient is based on the confusion matrix, which is an indication of consistency. To avoid any

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TABLE XI
COMPUTATIONAL TIME (SECONDS), FLOPS, AND PARAMETERS FOR SIX FSL-BASED METHODS

	Methods	DFSL+SVM [17]	DFSL+NN [17]	DCFSL [21]	CMFSL [23]	Gia-CFSL [32]	GCC-FSL
University of Pavia	Training time	1896.71s	834.76s	1482.50s	991.87s	2008.79s	3460.26s
	Testing time	6.25s	11.12s	6.23s	5.82s	4.06s	7.86s
	FLOPs	0.041135921G	0.041135921G	0.059785344G	0.244174752G	5.332537344G	4.2739648G
	Parameters	0.038140M	0.038140M	0.034920M	0.613976M	0.185192M	0.185192M
Houston	Training time	2645.99s	2729.16s	1338.27s	1053.13s	2947.27s	5317.03s
	Testing time	3.06s	4.51s	2.93s	1.21s	1.53s	2.65s
	FLOPs	0.041135921G	0.041135921G	0.059785344G	0.244174752G	5.332537344G	4.2739648G
	Parameters	0.038140M	0.038140M	0.034920M	0.613976M	0.185192M	0.185192M
Hangzhou	Training time	613.40s	558.84s	1212.54s	490.24s	1686.21s	1667.17s
	Testing time	11.91s	15.71s	19.33s	13.58s	14.44s	14.85s
	FLOPs	0.041135921G	0.041135921G	0.059785344G	0.244174752G	5.332537344G	4.2739648G
	Parameters	0.038140M	0.038140M	0.034920M	0.613976M	0.185192M	0.185192M

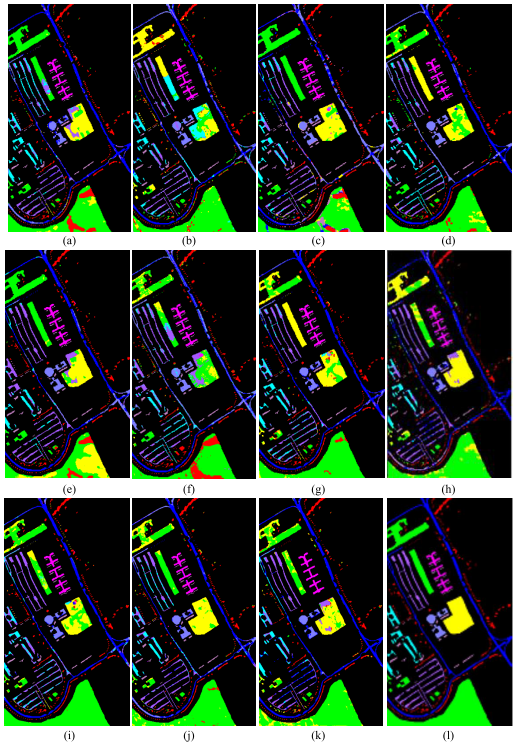


Fig. 11. Classification maps for the University of Pavia dataset. (a) SVM. (b) 3D-CNN. (c) SSRN. (d) MCNN. (e) SPRN. (f) DFSL + NN. (g) DFSL + SVM. (h) DCFSL. (i) CMFSL. (j) Gia-CFSL. (k) GCC-FSL. (l) Ground Truth.

bias induced by random sampling, we conducted ten trials and reported average results along with standard deviation. As shown in Tables VII–IX, traditional SVM has the lowest classification accuracy, which indicates that SVM is difficult to extract effective features from HSIs for classification. At the same time, when the number of labeled samples is small, the classification accuracy of SVM is also lower than that of other methods, which indicates that SVM cannot effectively solve the small-sample-size problem. The CNN-based methods, such as 3D-CNN, SSRN, MCNN, and SPRN, are discussed from a professional academic perspective. Among these methods, SSRN is a traditional 3-D convolution with a single-branch cascade that incorporates skip connections to form a residual network. MCNN follows a 2-D CNN with covariance pooling to extract second-order information from features. SPRN deploys parallel 2-D residual CNNs on multiple partitions of spectral bands, followed by element-wise convolutions for feature fusion. These deep-learning methods mentioned above outperform traditional SVM methods because discriminative and reliable features can be extracted to exploit deep representation and appropriate multilayer architectures are beneficial to effectively classify hyperspectral data. However, the lack of labeled samples is a major problem for CNN-based HSI classification methods due to the potential overfitting caused by the small number of labeled training samples. Fortunately, FSL methods, including DFSL + NN and DFSL + SVM, can achieve impressive classification accuracy with fewer labeled samples since transferable knowledge can be learned from source-domain data to help classify the unknown-class samples in the target domain. DCFSL, as a cross-domain FSL method, introduces a domain adaptive module that can mitigate the domain-shift problem, which can further improve classification results. CMFSL constructs a kind of class-covariance measure learning metric space, which improves the decision boundary of nonlinear classifier by considering the distribution of each class in the potential space, thus improving the classification accuracy with few-shot labeled samples. The Gia-FSL method, oriented FSL classification, mitigates the impact of domain shift by considering nonlocal spatial information with strong correspondence.

The proposed GCC-FSL method is also a cross-domain FSL method. It incorporates GC into CNN to extract graph-structure information of HSI and improves the similarity between similar samples by positive and negative pairs' construction module, in which sample distribution is achieved by distance computation of cross-domain data. It can be observed in Tables VII–IX that the proposed GCC-FSL produces optimal OA, AA, and κ compared to Gia-CFSL and other competing methods. For the University of Pavia dataset, OA, AA, and κ of GCC-FSL increased by 2.32%, 1.26%, and 2.89%, respectively, compared to that of Gia-CFSL. For the Houston dataset, OA, AA, and κ increased by 1.58%, 1.13%, and 1.73%, respectively, compared to the optimal comparable method of DCFSL. For the Hangzhou dataset, OA, AA, and κ of GCC-FSL increased by 2.26%, 2.17%, and 3.90% compared to that of Gia-CFSL, respectively.

2) Visual Comparison: Figs. 11–13 visualize the classification results of the best-trained networks from a professional academic perspective. It is evident that the SVM and four CNN-based methods (3D-CNN, SSRN, MCNN, and SPRN) generate classification maps with a significant amount of noise, while the FSL-based methods (DFSL + NN, DFSL + SVM, DCFSL, CMFSL, Gia-CFSL, and the proposed GCC-FSL) produce smoother maps, despite certain classes

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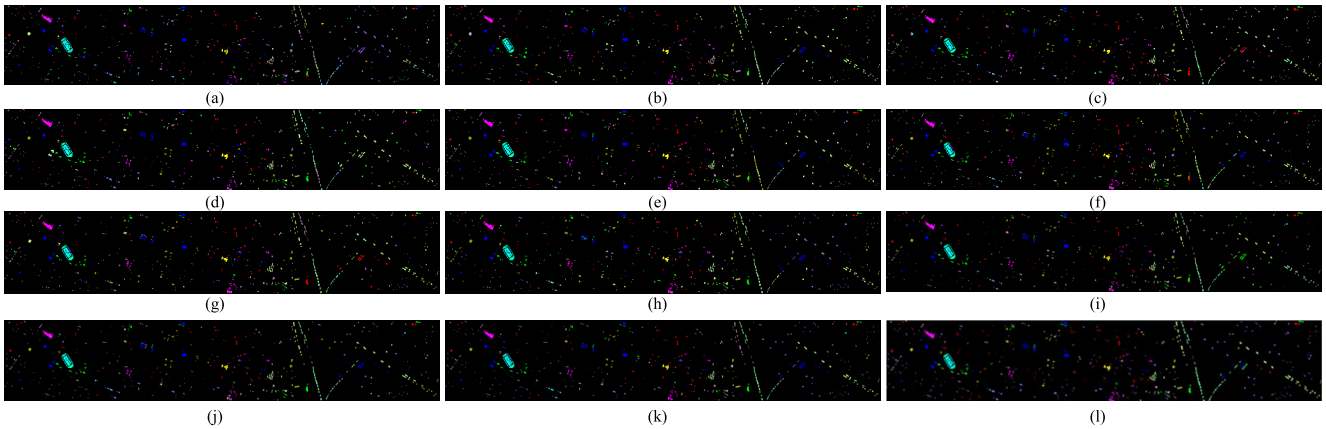


Fig. 12. Classification maps for the Houston dataset. (a) SVM. (b) 3D-CNN. (c) SSRN. (d) MCNN. (e) SPRN. (f) DFSL + NN. (g) DFSL + SVM. (h) DCFSL. (i) CMFSL. (j) Gia-CFSL. (k) GCC-FSL. (l) Ground truth.

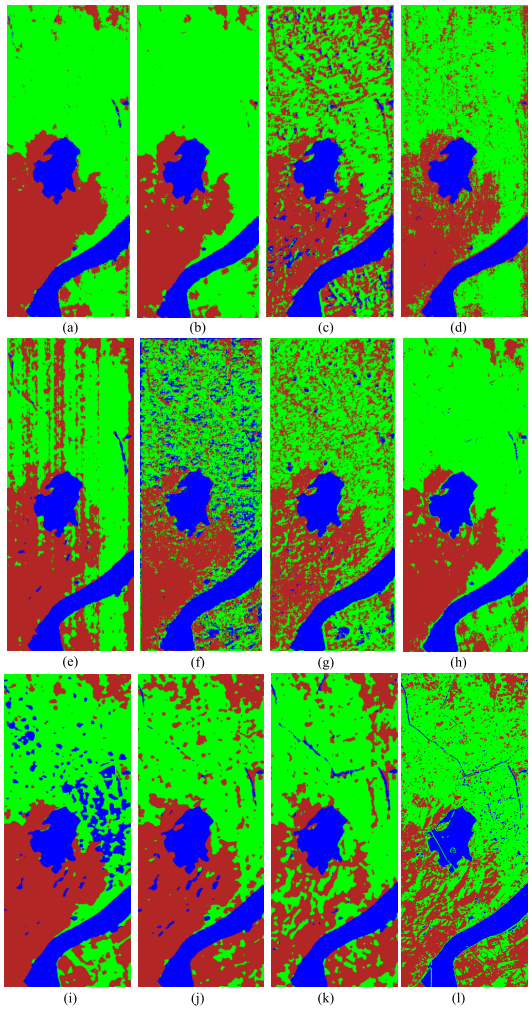


Fig. 13. Classification maps for the Hangzhou dataset. (a) SVM. (b) 3D-CNN. (c) SSRN. (d) MCNN. (e) SPRN. (f) DFSL + NN. (g) DFSL + SVM. (h) DCFSL. (i) CMFSL. (j) Gia-CFSL. (k) GCC-FSL. (l) Ground truth.

showing noticeable misclassifications. Compared to other FSL-based methods, GCC-FSL provides the most accurate and smoothest classification maps for all three datasets. Taking the Hangzhou dataset as an example, the distribution of *plants* and *land/building* is scattered, and their corresponding spatial regions are difficult to determine in detail. These factors hinder the performance of many classification networks. In contrast, the suggested GCC-FSL approach demonstrates significant superiority over other networks in handling this type, yielding impressive classification performance.

B. Ablation Experiments

A battery of ablation experiments are conducted to verify various facets of the GCC-FSL design. In the proposed network, the similarity between positive pairs increases and the similarity between negative pairs decreases by using the proposed GCC module, which induces distributed alignment of the samples in the source and target feature distribution spaces in order to minimize the effect of domain shift. We design an additional network to compare with the proposed GCC-FSL as shown in Table X to measure the effect of the GCC module in GCC-FSL. These networks are as follows.

1) *Network A:* The proposed network without GCC module.
2) *Network B:* The proposed GCC-FSL network.

The results of networks A and B are evaluated in terms of OA, AA, κ, parameters, and floating-point operations (FLOPs) in Table X. The proposed GCC-FSL stands out demonstrating that the GCC module helps to improve the classification performance of the proposed GCC-FSL.

C. Stability Analysis

In order to analyze the influence of different numbers of labeled training samples on classification performance of various FSL methods, we randomly select labeled samples from the range of $[1, 2, 3, 4, 5]$ and the classification results for three datasets are reported in Fig. 14, from which we can find that the OAs of various methods usually increase as the number of labeled increasing and the GCC-FSL always achieves higher classification accuracy and better stability even with excessive limited training samples (e.g., only one training sample per class).

D. Computational Complexity

In order to demonstrate the computational efficiency of mentioned FSL-based methods (DFSL + NN, DFSL + SVM,

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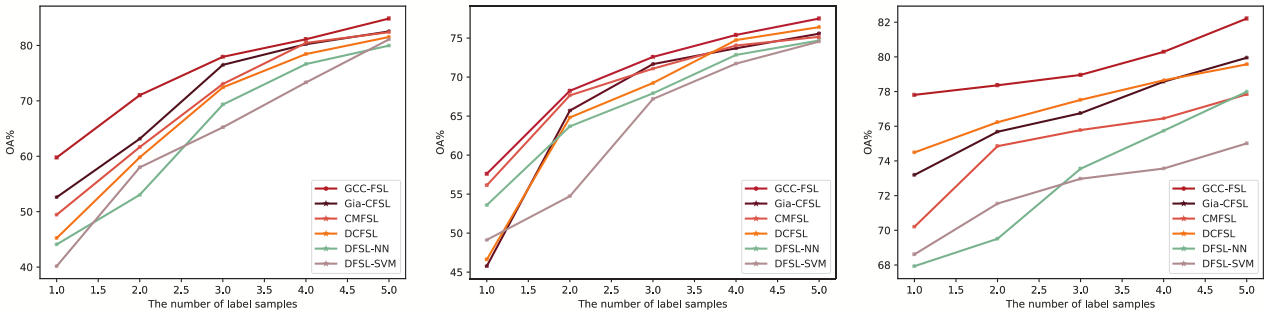


Fig. 14. Classification accuracy (%) with different numbers of support samples per class. (a) University of Pavia. (b) Houston. (c) Hangzhou.

DCFSL, and GCC-FSL), training time, testing time, the number of parameters, and FLOPs for three datasets are listed in Table XI. All results are obtained under the situation that five labeled samples per class are selected randomly. The training time is mainly consumed by transfer learning and the testing time is mainly predicting time of unlabeled samples for each HSI data set. The FLOPs and parameters were investigated to evaluate the computational complexity of deep-learning methods. From Table IX, it can be observed that the computational time, FLOPs, and the number of parameters of the proposed method are all acceptable, compared with that of other methods. All experiments are conducted on a system with two NVIDIA Geforce RTX 2080Ti GPUs. Implementation is in Pytorch on Ubuntu 19.04. The memory is 64 GB.

VI. Conclusion

In this article, we propose GCC-FSL for cross-domain HSI classification. The proposed algorithm is composed of feature extraction network, positive and negative pairs' construction module, and GCC module. To improve feature representation ability, spatial–spectral features are extracted by a 3D-CNN network with residual structure. By using class prototypes of source domain and target domain, the positive and negative pairs can be constructed to contribute cross-domain data distribution, which can effectively alleviate the domain-shift. Most important of all, a GCC module is proposed to construct graph-contrast loss, by which the similarity of positive and negative pairs is adjusted to further solve the domain-shift problem and the global graph-structure information is explored to enhance the ability of feature expression. Experiments are conducted on four widely used HSI datasets, including one source-domain dataset and three target-domain datasets. The experimental results demonstrate that the proposed GCC-FSL not only outperforms several state-of-the-art supervised deep-learning methods when few labeled samples are available but also more competitive than recently proposed cross-domain FSL methods for HSI classification.

References

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] Z. Ye, S. Prasad, W. Li, J. E. Fowler, and M. He, "Classification based on 3-D DWT and decision fusion for hyperspectral image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 173–177, Jan. 2014.

[3] Y.-N. Liu et al., "The advanced hyperspectral imager: Aboard China's GaoFen-5 satellite," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 4, pp. 23–32, Dec. 2019.

[4] C. M. Gevaert, J. Suomalainen, J. Tang, and L. Kooistra, "Generation of spectral–temporal response surfaces by combining multispectral satellite and hyperspectral UAV imagery for precision agriculture applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3140–3146, Jun. 2015.

[5] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[6] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[7] V. Slavkovikj, S. Verstockt, W. De Neve, S. Van Hoecke, and R. Van de Walle, "Hyperspectral image classification with convolutional neural networks," in *Proc. 23rd ACM Int. Conf. Multimedia.*, 2015, pp. 1159–1162.

[8] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[9] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[10] J. Zheng, Y. Feng, C. Bai, and J. Zhang, "Hyperspectral image classification using mixed convolutions and covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 522–534, Jan. 2021.

[11] Y. Li, H. Zhang, and Q. Shen, "Spectral–Spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[13] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[14] X. Zhang, S. Shang, X. Tang, J. Feng, and L. Jiao, "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5507714.

[15] S. Ravi and H. Larochelle, "Optimization as a model for few-shot learning," in *Proc. Int. Conf. Learn. Represent.*, 2016, pp. 1–11.

[16] Q. Chi, G. Lv, G. Zhao, and X. Dong, "A novel knowledge distillation method for self-supervised hyperspectral image classification," *Remote Sens.*, vol. 14, no. 18, p. 4523, Sep. 2022.

[17] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[18] L. Bruzzone and D. F. Prieto, "Unsupervised retraining of a maximum likelihood classifier for the analysis of multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 2, pp. 456–460, 2001.

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[19] T. Han and D. G. Goodenough, "Investigation of nonlinearity in hyperspectral remotely sensed imagery—A nonlinear time series analysis approach," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2007, pp. 1556–1560.

[20] M. Hesketh and G. A. Sánchez-Azofeifa, "The effect of seasonal spectral variation on species classification in the panamanian tropical forest," *Remote Sens. Environ.*, vol. 118, pp. 73–82, Mar. 2012.

[21] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022, Art. no. 5501618.

[22] W. Wang, F. Liu, J. Liu, and L. Xiao, "Cross-domain few-shot hyperspectral image classification with class-wise attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023.

[23] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[24] S. Arora, H. Khandeparkar, M. Khodak, O. Plevrakis, and N. Saunshi, "A theoretical analysis of contrastive unsupervised representation learning," 2019, *arXiv:1902.09229*.

[25] X. Huang, M. Dong, J. Li, and X. Guo, "A 3-D-swin transformer-based hierarchical contrastive learning method for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5411415.

[26] S. Zhang, Z. Chen, D. Wang, and Z. J. Wang, "Cross-domain few-shot contrastive learning for hyperspectral images classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[27] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5521213.

[28] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 5506214.

[29] Z. Ye, T. Sun, S. Shi, L. Bai, and J. E. Fowler, "Local–global active learning based on a graph convolutional network for semi-supervised classification of hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[30] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.

[31] Y. Zhang, W. Li, M. Zhang, and R. Tao, "Dual graph cross-domain few-shot learning for hyperspectral image classification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 3573–3577.

[32] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, pp. 1–14, Jun. 2022.

[33] S. Zhang, H. Tong, J. Xu, and R. Maciejewski, "Graph convolutional networks: A comprehensive review," *Comput. Social Netw.*, vol. 6, no. 1, pp. 1–23, Dec. 2019.

[34] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[35] V. Menon, S. Prasad, and J. E. Fowler, "Hyperspectral classification using a composite kernel driven by nearest-neighbor spatial features," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 2100–2104.

[36] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4080–4090.

[37] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[38] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. Int. Conf. Artif. Intell. Statist. (AISTATS)*, in JMLR Workshop and Conference Proceedings, 2010, pp. 249–256.

[39] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[40] S. Zhong, C.-I. Chang, and Y. Zhang, "Iterative support vector machine for hyperspectral image classification," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3309–3312.

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